

Zaid Asaad

Lahore, Pakistan · zaidasaad385@gmail.com · +923208669499
www.linkedin.com/in/zaid-asaad · <https://github.com/Zaid385>

Introduction

6th semester Computer Science student at COMSATS University with experience in full-stack web development, real-time networked systems, and audio software. Proficient in Node.js, Python, TCP socket programming, REST API design, and MongoDB. Passionate about backend engineering, networking protocols, and audio/DSP systems.

Education

Bachelor of Science in Computer Science COMSATS University Islamabad, Lahore Campus	2024 – Present
Intermediate (HSSC) Punjab College of Science	2021 – 2023
Matriculation (SSC) Lahore Grammar School	2019 – 2021

Technical Skills

Languages: C, Python, TypeScript, JavaScript, Java, SQL
Frameworks & Libraries: React, Node.js, Express.js, JavaFX, PySide6, Tone.js, Vite.js
Backend: REST API Design, JWT, Session Auth, bcrypt, AES/Fernet Encryption
Databases: MongoDB (PyMongo, Mongoose), IndexedDB (Dexie)
Networking: Raw TCP Sockets, Binary Protocol Design, Multi-threaded Concurrency
Cloud & Tools: Cloudinary, Render, Railway, Git, Docker, npm

Projects

Reson — Browser-Based Music Production App github.com/Zaid385/Reson ·

Live: reson-4nav.onrender.com

Stack: React 19, TypeScript, Tone.js, Zustand, Dexie, Vite

- Built a client-side audio engine on Tone.js supporting velocity-sensitive sample playback across a 32-pad grid, with a built-in sample browser and waveform editor for trimming/chopping.
- Integrated the Web MIDI API to map external MIDI controller input directly to pad triggers, translating note and velocity data in real time for hardware-driven performance.
- Persisted all projects locally via IndexedDB (Dexie) with a fully offline-capable PWA build, requiring no backend server.

AudioFlow — Full-Stack Music Streaming Web Application github.com/Zaid385/AudioFlow ·

Live: audioflow-4pg4.onrender.com

Stack: TypeScript, Node.js, Express.js, MongoDB, Cloudinary, JWT

- Built a dual-interface backend serving server-rendered EJS views (session auth) and a stateless REST API (JWT) from a single Express.js router.
- Implemented role-based access control (customer/admin) and a Cloudinary media pipeline with Multer and audio metadata parsing for a paginated, filterable song catalog.

Tether — Real-Time Messaging Application github.com/Zaid385/Tether

Stack: Python, PySide6, MongoDB, TCP Sockets, bcrypt, Fernet

- Designed a custom JSON-over-TCP protocol with length-prefixed framing supporting 20+ packet types across auth, messaging, presence, and file transfer.
- Built a multi-threaded server with per-client threading and a `PacketDispatcher` routing packets to service handlers; secured auth with bcrypt and AES-128 Fernet.

JavaFX Music Player — Desktop Audio Application github.com/Zaid385/Music-Player

Stack: Java 23, JavaFX 24, FXML, SceneBuilder

- Designed a layered OOP architecture (`Player`, `PlaylistManager`, `Login`, `Register`) with clean scene-switching and live metadata playback via `MediaPlayer` callbacks.